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20 General References

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21 Authors

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22 Date of Revision

18 February 2009.

Adipic Acid

1 Nonproprietary Names

PhEur: Adipic Acid

USP-NF: Adipic Acid

2 Synonyms

Acidum adipicum; acifloctin; acinetten; adilactetten; asapic; 1,4-butanedicarboxylic acid; E355; 1,6-hexanedioic acid; *Inipol DS*.

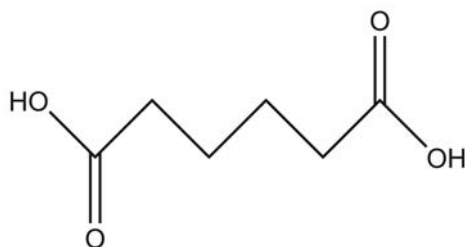
3 Chemical Name and CAS Registry Number

Hexanedioic acid [124-04-9]

4 Empirical Formula and Molecular Weight

C₆H₁₀O₄ 146.14

5 Structural Formula



6 Functional Category

Acidifying agent; buffering agent; flavoring agent.

7 Applications in Pharmaceutical Formulation or Technology

Adipic acid is used as an acidifying and buffering agent in intramuscular, intravenous and vaginal formulations. It is also used in food products as a leavening, pH-controlling, or flavoring agent.

Adipic acid has been incorporated into controlled-release formulation matrix tablets to obtain pH-independent release for both weakly basic^(1,2) and weakly acidic drugs.^(3,4) It has also been incorporated into the polymeric coating of hydrophilic monolithic systems to modulate the intragel pH, resulting in zero-order release

of a hydrophilic drug.⁽⁵⁾ The disintegration at intestinal pH of the enteric polymer shellac has been reported to improve when adipic acid was used as a pore-forming agent without affecting release in the acidic media.⁽⁶⁾ Other controlled-release formulations have included adipic acid with the intention of obtaining a late-burst release profile.⁽⁷⁾

8 Description

Adipic acid occurs as a white or almost white, odorless nonhygroscopic crystalline powder. The crystal structure of adipic acid is monoclinic holohedral.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for adipic acid.

Test	PhEur 6.0	USP32-NF27
Identification	+	+
Characters	+	—
Melting range	151–154°C	151–154°C
Appearance of solution	+	—
Loss on drying	≤ 0.2%	≤ 0.2%
Residue on ignition	—	≤ 0.1%
Sulfated ash	≤ 0.1%	—
Chlorides	≤ 200 ppm	≤ 0.02%
Nitrates	≤ 30 ppm	≤ 0.003%
Sulfates	≤ 500 ppm	≤ 0.05%
Iron	≤ 10 ppm	≤ 0.001%
Heavy metals	≤ 10 ppm	≤ 0.001%
Related substances	+	—
Assay (anhydrous)	99.0–101.0%	99.0–101.0%

10 Typical Properties

Acidity/alkalinity pH = 2.7 (saturated solution at 25°C); pH = 3.2 (0.1% w/v aqueous solution at 25°C)

Boiling point 337.5°C

Dissociation constant

pK_{a1}: 4.418 at 25°C;

pK_{a2}: 5.412 at 25°C.

Density 1.360 g/cm³

Flash point 196°C (closed cup)

Heat of combustion 17 653.9 kJ/mol (4219.28 kcal/mol) at 25°C
Heat of solution 33.193 kJ/mol (7.9 kcal/mol) at 25°C
Melting point 152.1°C
Solubility see Table II.
Vapor pressure 133.3 Pa (1 mmHg) at 159.5°C
Viscosity (dynamic) 4.54 mPa s (4.54 cP) at 160°C for molten adipic acid.

Table II: Solubility of adipic acid.

Solvent	Solubility at 20°C unless otherwise stated
Acetone	Soluble
Benzene	Practically insoluble
Cyclohexane	Slightly soluble
Ethanol (95%)	Freely soluble
Ether	1 in 157.8 at 19°C
Ethyl acetate	Soluble
Methanol	Freely soluble
Petroleum ether	Practically insoluble
Water	1 in 71.4 1 in 0.6 at 100°C

11 Stability and Storage Conditions

Adipic acid is normally stable but decomposes above boiling point. It should be stored in a tightly closed container in a cool, dry place, and should be kept away from heat, sparks, and open flame.

12 Incompatibilities

Adipic acid is incompatible with strong oxidizing agents as well as strong bases and reducing agents. Contact with alcohols, glycols, aldehydes, epoxides, or other polymerizing compounds can result in violent reactions.

13 Method of Manufacture

Adipic acid is prepared by nitric acid oxidation of cyclohexanol or cyclohexanone or a mixture of the two compounds. Recently, oxidation of cyclohexene with 30% aqueous hydrogen peroxide under organic solvent- and halide-free conditions has been proposed as an environmentally friendly alternative for obtaining colorless crystalline adipic acid.⁽⁸⁾

14 Safety

Adipic acid is used in pharmaceutical formulations and food products. The pure form of adipic acid is toxic by the IP route, and moderately toxic by other routes. It is a severe eye irritant, and may cause occupational asthma.

LD₅₀ (mouse, IP): 0.28 g/kg⁽⁹⁾

LD₅₀ (mouse, IV): 0.68 g/kg

LD₅₀ (mouse, oral): 1.9 g/kg

LD₅₀ (rat, IP): 0.28 g/kg

LD₅₀ (rat, oral): >11 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled.

Adipic acid is combustible and can react with oxidizing materials when exposed to heat and flame. It emits acrid smoke and fumes

when heated to decomposition. Dust explosion is possible if in powder or granular form, mixed with air.

Adipic acid irritates the eyes and respiratory tract. Protective equipment such as respirators, safety goggles, and heavy rubber gloves should be worn when handling adipic acid.

16 Regulatory Status

GRAS listed. Included in the FDA Inactive Ingredients Database (IM, IV, and vaginal preparations). Accepted for use as a food additive in Europe. Included in an oral pastille formulation available in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.

17 Related Substances

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18 Comments

A specification for adipic acid is contained in the Food Chemicals Codex (FCC).⁽¹⁰⁾

The EINECS number for adipic acid is 204-673-3. The PubChem Compound ID (CID) for adipic acid is 196.

19 Specific References

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20 General References

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21 Author

D Law.

22 Date of Revision

27 February 2009.

1 Nonproprietary Names

JP: Agar
PhEur: Agar
USP-NF: Agar

2 Synonyms

Agar-agar; agar-agar flake; agar-agar gum; Bengal gelatin; Bengal gum; Bengal isinglass; Ceylon isinglass; Chinese isinglass; E406; gelosa; gelose; Japan agar; Japan isinglass; layor carang.

3 Chemical Name and CAS Registry Number

Agar [9002-18-0]

4 Empirical Formula and Molecular Weight

See Section 5.

5 Structural Formula

Agar is a dried, hydrophilic, colloidal polysaccharide complex extracted from the agarocytes of algae of the Rhodophyceae. The structure is believed to be a complex range of polysaccharide chains having alternating α -(1 $\rightarrow$ 3) and β -(1 $\rightarrow$ 4) linkages. There are three extremes of structure noted: namely neutral agarose; pyruvated agarose having little sulfation; and a sulfated galactan. Agar can be separated into a natural gelling fraction, agarose, and a sulfated nongelling fraction, agarpectin.

6 Functional Category

Emulsifying agent; stabilizing agent; suppository base; suspending agent; sustained-release agent; tablet binder; thickening agent; viscosity-increasing agent.

7 Applications in Pharmaceutical Formulation or Technology

Agar is widely used in food applications as a stabilizing agent. In pharmaceutical applications, agar is used in a handful of oral tablet and topical formulations. It has also been investigated in a number of experimental pharmaceutical applications including as a sustained-release agent in gels, beads, microspheres, and tablets.⁽¹⁻⁴⁾ It has also been reported to work as a disintegrant in tablets.⁽⁵⁾ Agar has been used in a floating controlled-release tablet; the buoyancy in part being attributed to air entrapped in the agar gel network.⁽⁶⁾ It can be used as a viscosity-increasing agent in aqueous systems. Agar can also be used as a base for nonmelting, and nondisintegrating suppositories.⁽⁷⁾ Agar has an application as a suspending agent in pharmaceutical suspensions.⁽⁸⁾

8 Description

Agar occurs as transparent, odorless, tasteless strips or as a coarse or fine powder. It may be weak yellowish-orange, yellowish-gray to pale-yellow colored, or colorless. Agar is tough when damp, brittle when dry.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for agar.

Test	JP XV	PhEur 6.3	USP32-NF27
Identification	+	+	+
Characters	+	+	—
Swelling index	—	+	—
Arsenic	—	—	≤ 3 ppm
Lead	—	—	$\leq 0.001\%$
Sulfuric acid	+	—	—
Sulfurous acid and starch	+	—	—
Gelatin	—	+	+
Heavy metals	—	—	$\leq 0.004\%$
Insoluble matter	≤ 15.0 mg	$\leq 1.0\%$	≤ 15.0 mg
Water absorption	≤ 75 mL	—	≤ 75 mL
Loss on drying	$\leq 22.0\%$	$\leq 20.0\%$	$\leq 20.0\%$
Microbial contamination	—	$\leq 10^3$ cfu/g ^(a)	+
Total ash	$\leq 4.5\%$	$\leq 5.0\%$	$\leq 6.5\%$
Acid-insoluble ash	$\leq 0.5\%$	—	$\leq 0.5\%$
Foreign organic matter	—	—	$\leq 1.0\%$
Limit of foreign starch	—	—	+

(a) Total viable aerobic count, determined by plate-count.

10 Typical Properties

NIR spectra see Figure 1.

Solubility Soluble in boiling water to form a viscous solution; practically insoluble in ethanol (95%), and cold water. A 1% w/v aqueous solution forms a stiff jelly on cooling.

11 Stability and Storage Conditions

Agar solutions are most stable at pH 4–10.

Agar should be stored in a cool, dry, place. Containers of this material may be hazardous when empty since they retain product residues (dust, solids).

12 Incompatibilities

Agar is incompatible with strong oxidizing agents. Agar is dehydrated and precipitated from solution by ethanol (95%). Tannic acid causes precipitation; electrolytes cause partial dehydration and decrease in viscosity of sols.⁽⁹⁾

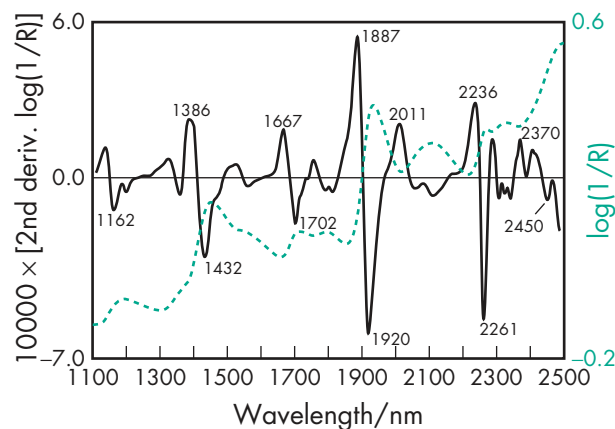


Figure 1: Near-infrared spectrum of agar measured by reflectance.